KARTHIK SUBRAMANIAN

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EDUCATION

University of California, Berkeley *May 2029*
B.S. in Electrical Engineering and Computer Sciences | Regents' and Chancellor's Scholar

PROFESSIONAL EXPERIENCE

Mosi Audio **Santa Clara, CA**
Software Development Intern *May 2024 - Present*
- Collaborated with the CEO to streamline Python workflows, automating unit tests & reducing manual testing time by 30%.
- Delivered firmware enhancements and built an encryption/logging framework for an electronic music creation platform.

Computer Science Undergraduate Association **Berkeley, CA**
Technology Committee Officer *Oct 2025 - Present*
- Maintained Linux servers, LDAP, and email systems; supported internal tools like a Discord bot and CV GeoGuessr app.
- Held office hours to assist students with Git/GitHub and Python/Java; coordinated talks, workshops, and industry events.

Outer Rim Exploration **Berkeley, CA**
Software Engineering and Embedded Systems Intern *Sep 2025 - Jan 2026*
- Assembled and deployed hodoscopes for muography projects, integrating IoT devices and sensors (SiPMs, gyroscopes).
- Developed real-time C++ firmware and Python pipelines for data post-processing, calibration, and simulation integration.

Space Technologies at California **Berkeley, CA**
Software Engineer *Sep 2025 - Jan 2026*
- Built computer vision tools for a lunar hopper, merging camera feeds, refining imagery, and ensuring precise alignment.
- Enabled autonomous navigation, crater avoidance, and scientific data collection, supporting detailed lunar resource maps.

George Mason University **Remote**
Research Intern *Feb 2024 - Dec 2024*
- Researched coronal mass ejections under Prof. Poland, leveraging AI/ML techniques to analyze supersonic flow data.
- Published in the Journal of Student-Scientists' Research; Presented at the 2024 American Geophysical Union Conference.

FRC Skywalkers **Fremont, CA**
Software Lead, Operations Lead, Business and Outreach Captain *Jun 2021 - Jun 2025*
- Secured $60K+ annually in sponsorships from Apple, NASA, BAE Systems, NVIDIA, Albertsons, PG&E, and others.
- Led robotics and Python programming workshops for 3,000+ students at libraries, schools, nonprofits, and STEM fairs.
- Mentored 8 FRC, FTC, and FLL teams and supported 14 additional local teams with technical and financial guidance.

LEADERSHIP EXPERIENCE

Student Bono **Fremont, CA**
Founder and Head of Chapter Network, Chief Expansion Officer, Nonprofit Coordination Director *Feb 2022 - Jun 2025*
- Established chapter clubs in over 25 high schools across the nation, 9 international chapters (Egypt, France, India, Indonesia, Morocco, Pakistan, the Philippines, Spain, and Ukraine), mobilizing over 1,800 student interns worldwide.
- Partnered with over 40 non-profits, including StreetCode Academy, the Wounded Warrior Project, and Rainforest Action Network, to help educate 500k+ students and fundraise $75k+ for educational, environmental, and humanitarian causes.
- Directed the largest global K-12 VR study (160 students, 12 countries) with the Stanford Graduate School of Education.

American High School Engineering and Technology Club **Fremont, CA**
President, Head of AI and Computational Statistics Branches *Oct 2021 - Jun 2025*
- Led the largest club on campus with 330+ active members; managed operations, recruitment, and event planning.
- Launched a speaker series featuring content creators, researchers, and industry experts, generating 3K+ YouTube views
- Encompasses 10 branches, including FTC & VEX Robotics, American Rocketry Challenge, and Tech Student Association.

Eagle Explorer **Fremont, CA**
Founder, President, Editor-in-Chief *Oct 2021 - Jun 2025*
- Founded the school's first STEAM publication, publishing recurring online issues on diverse STEM topics.
- Led a 13-member editorial team of writers, editors, and designers to publish high-quality technical content.

HONORS & AWARDS

Invention Convention
- SafeBox (Keeps package deliveries safe, 2021): *Finalist at National Invention Convention; Broadcom MASTERS nominee*
- CalmSpace (Assists caregivers in managing autism meltdowns, 2022): *Merit Award at California Invention Convention*
- AquaMate (IoT device to track water intake, 2023): *Top Startup Pitch at LaunchX, Finalist at Invention Convention U.S. Nationals, National Finalist & Honorable Mention at University of Iowa Jacobson Institute's Innovator Competition*
- Supernova Classification (ML-based classifier, 2024): *Merit Award at California Invention Convention*

Quiz Bowl (QB) + History Bowl (HB)
- Co-founded and led the school's 5 QB and HB teams; Organized 8 scrimmages and the East Bay Quiz Bowl Tournament.
- 2024 and 2025 national championship qualifier at National Academic Quiz Tournaments; 1st in California Cup Series.
- National History Bowl and International History Olympiad Qualifier; 2nd Place at History Bowl Western Regional.

National History Day
- Created 7 original, research-based websites, integrating archival sources, interactive design, and historical analysis.
- California Statewide Winner & National Qualifier; 6x California State Finalist; Special Awards from the California Office of Historic Preservation, the California Historical Society, and the California Council for the Promotion of History.

Distinctions: National Qualifier at AIME; 4x National Science Bee Qualifier; 3x National History Bee Qualifier; 2x Team USA Representative at the Geography Bee World Championships; 2x International Astronomy and Astrophysics Competition Finalist

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Skills: Python, C/C++, Java, JavaScript, MATLAB, SQL, Git, React, Node.js, Arduino, ML (TensorFlow, PyTorch), Data Analysis
Interests: Writing, Board Games, Stargazing, Philosophy, Travel, Political Activism & Civic Engagement, Film & TV, Geoguessr